CSCE 5612 Embedded Hardware/Software Design – Spring 2026

Project Milestone One: System Design and Hardware Justification

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Udayasai Ramineni (11802872)

Ashfaaq Ahamed Kapatrala(11786442)

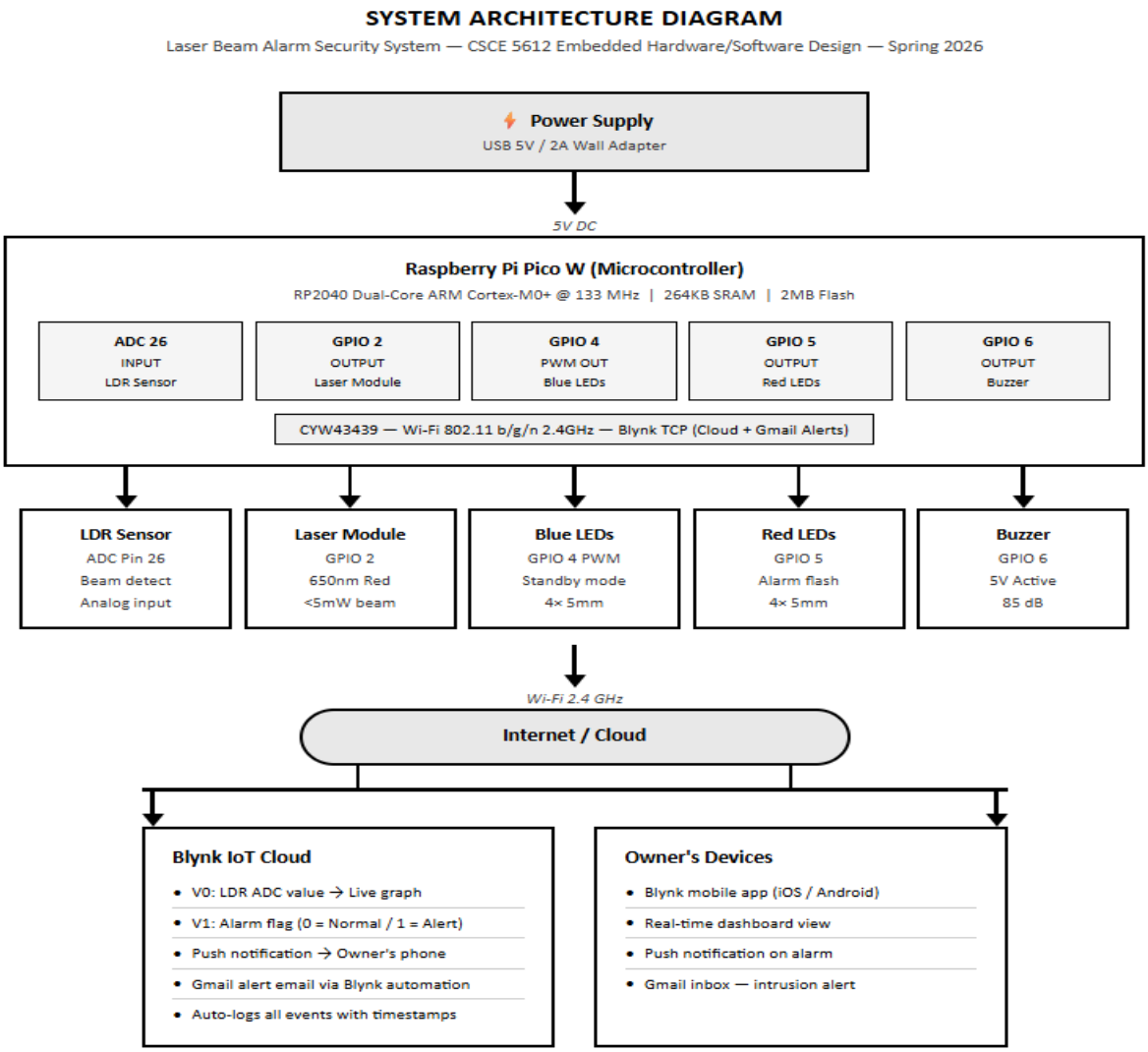
Project Title

Laser Beam Alarm Security System with Blynk Cloud Monitoring & Gmail Alerts

Motivation

Physical security devices are essential security tools for protecting doorways, gates, and restricted areas. This project builds on a conventional embedded system implementation of a laser beam trip wire alarm system. This project has been extended to include Internet of Things (IoT) features such as real-time monitoring of sensors via the Blynk cloud platform and sending Gmail alerts on intrusion detection. This extension of the embedded system to include IoT features aligns well with modern IoT and embedded system design principles.

1. System Architecture Diagram



Input : Laser module, LDR sensor

Processing : Raspberry Pi Pico W (RP2040 microcontroller)

Output : Buzzer, LEDs (local), Blynk dashboard (cloud), Gmail alert (cloud)

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2. Hardware Component Selection

2.1 Raspberry Pi Pico W (Microcontroller)

Attribute	Detail
Microcontroller	RP2040 dual-core ARM Cortex-M0+ @ 133 MHz
Operating Voltage	3.3V I/O, 5V USB power input
Flash / RAM	2 MB Flash, 264 KB SRAM
Communication	Wi-Fi 802.11 b/g/n (2.4 GHz) via CYW43439
ADC	3× 12-bit ADC channels
GPIO	26 multi-function GPIO pins
Programming	MicroPython / C SDK
Power Consumption	~100 mA active (Wi-Fi on)

Justification: The Pico W is significant because it has the same RP2040 microcontroller as the Pico but has a Wi-Fi module. This Wi-Fi is essential to connect to the Blynk cloud and send Gmail SMTP. It is cost-effective, has wide support for MicroPython, and has enough analog inputs and GPIOs to connect to all sensors and actuators.

URL: https://www.adafruit.com/product/5526

2.2 Laser Module (KY-008 or equivalent)

Attribute	Detail
Operating Voltage	5V DC
Output Wavelength	650 nm (red laser)
Output Power	< 5 mW (Class IIIA)
Current Draw	~30 mA
Interface	Digital GPIO (ON/OFF)

Justification: The laser module emits a tight beam directly at the LDR. It provides a steady and reliable light source regardless of the light around it, which makes it a more reliable option than other methods that use light.

URL: https://www.amazon.com/dp/B01MAWT5JX

2.3 LDR Sensor (Light Dependent Resistor)

Attribute	Detail
Resistance (dark)	~1 MΩ
Resistance (light)	~1–10 kΩ
Operating Voltage	3.3V – 5V
Interface	Analog (ADC)
Response Time	~20 ms

Justification: The LDR is used as a beam break detector. As the laser beam falls on the LDR, the resistance is reduced, and a low value is recorded by the ADC. As the beam is interrupted, the resistance increases sharply, causing the ADC value to exceed the threshold.

URL: https://www.adafruit.com/product/161

2.4 5V Active Buzzer

Attribute	Detail
Operating Voltage	5V DC
Current Draw	~30–40 mA
Sound Level	~85 dB
Interface	Digital GPIO (via transistor)
Frequency	Fixed internal oscillator

Justification: The system also sends an instant local audible alert once the intrusion is detected. The active buzzer requires a simple HIGH or LOW signal; no complex PWM tone generation is required. This makes the code easier.

URL: https://www.amazon.com/dp/B0716GH3QD

2.5 LEDs (4× Red, 4× Blue, 5mm)

Attribute	Detail
Operating Voltage	~2.0–3.4V forward voltage
Current Draw	10–20 mA each
Interface	GPIO (with current-limiting resistor)
Color	Red (alarm indicator), Blue (standby)

Justification: To display the status, the blue LED is kept steady during normal conditions. The red LED blinks during alarm conditions. The current is kept under control using resistors.

URL: https://www.amazon.com/dp/B01AUI4VQU

2.6 D400 NPN Transistors (×2)

Attribute	Detail
Type	NPN BJT
V_CE max	25V
I_C max	500 mA
Interface	GPIO base drive

Justification: A transistor-based switch was utilized to control the buzzer and the LED arrays, since these can source a larger amount of current than the Pico W GPIO pins can supply (which is a maximum of 12 mA per pin). The transistors act as current amplifiers and switches.

URL: https://www.amazon.com/dp/B07T5QRNXD

2.7 Resistors

Value	Quantity	Purpose
180 Ω	2	LED current limiting
1 kΩ	2	Transistor base resistor
10 kΩ	1	LDR pull-down voltage divider

URL: https://www.amazon.com/dp/B08FD1XVL6

2.8 PCB (Custom Gate-Shaped, via JLCPCB)

Justification: A custom PCB design eliminates the necessity for a breadboard, provides you with reliable and hard connections, and provides a neat and installable package with a gate-like shape for easy installation in a door or entryway.

URL: https://jlcpcb.com

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3. Power Consumption Analysis

Component	Voltage	Current	Power
Raspberry Pi Pico W (Wi-Fi active)	3.3V	~100 mA	~330 mW
Laser Module	5V	~30 mA	~150 mW
Buzzer (when active)	5V	~35 mA	~175 mW
LEDs × 4 (when on)	~2V	~10 mA each	~80 mW
LDR + resistor divider	3.3V	~0.3 mA	~1 mW
Total (worst case)		**~175 mA avg**	**~736 mW**

Power Source Justification: This system uses a USB 5V power source, which can be obtained from a USB adapter or a computer. This is a fixed security system that is always powered, and a USB adapter that supplies 5V and 2A is more than enough. Battery operation was not considered, as the constant use of Wi-Fi would discharge the battery in a matter of hours, and this is a 24/7 security system that is always online.Power Efficiency Strategies:

- In the absence of intrusion, the Pico W will check the LDR value every 100 ms but will send information to Blynk only after 2 seconds, which reduces Wi-Fi activity.
- LED control is software-based and uses PWM, which means that the LED is controlled at a lower duty cycle in standby mode.
- Deep sleep is not used, as the system is always online.

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4. Communication and Data Flow

4.1 Local Communication

- The Pico W speaks to nearby sensors and actuators using the GPIO and ADC pins on the microcontroller itself. There is no requirement for any external bus protocol to communicate with the sensors and actuators.

4.2 Wi-Fi / Cloud Communication

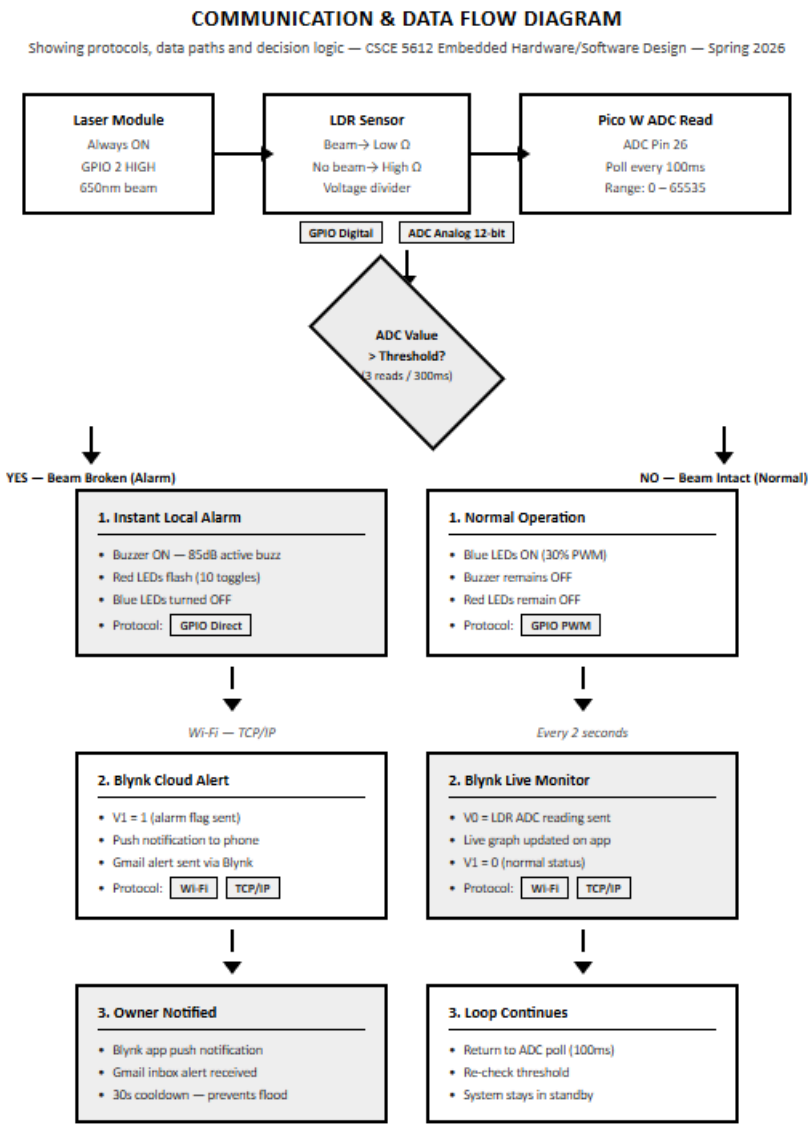
- Protocol: 802.11 b/g/n at 2.4 GHz
- TCP/IP Protocol Stack: CYW43439 chip

MicroPython
Blynk Cloud Integration

- The Pico W connects to the Blynk Cloud IoT platform using the MicroPython Blynk library over a TCP/IP network using ports 80/443.
    - Virtual Pins
    - V0: Continuous LDR ADC Reading (sent every 2 seconds to update the real-time graph on the Blynk app).
    - V1: Alarm flag (0 = normal, 1 = intrusion detected).
    - Blynk app and web interface show real-time data and also sends push notifications to the user's phone.
- Gmail Alert Integration

- When the intrusion is detected, the Pico W connects to the Gmail server using the MicroPython library over a TCP/IP network using port 465 and the SSL protocol.
    - An email is sent to the user using their app-specific password (Google 2FA + App Password).
    - The email includes a timestamp, the message "Laser beam broken — Intruder detected," and the raw LDR ADC reading at the time of intrusion.
    - A 30-second cooldown prevents multiple emails in the case of a sustained beam interruption.

4.1 Data Flow Summary



4.2 Data Logging

All the values of the virtual pins, along with the timestamps, get recorded automatically by the Blynk cloud. This is a passive way of storing the values received from the LDR sensor and the alarm pins. It is possible to check the values at any time using the Blynk app.

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5. System Integration Challenges and Solutions

Challenge 1: Wi-Fi connectivity latency when alarm hits

Problem: Connecting to Wi-Fi and sending out the SMTP email at the exact time of the alarm causes a 2-5 second delay before the notification occurs in the cloud.

Solution: The Pico W maintains a Wi-Fi connection from the time it is switched on. Blynk uses a persistent TCP socket for connectivity. So, the alarm is sent immediately. Gmail's SMTP is then opened after the local alarm (buzzer and LED) has gone off immediately.

Challenge 2: LDR sensitivity to ambient light

Problem: Changes in ambient light (e.g., sunlight or lights turning on) could change the ADC threshold and lead to false alarms.

Solution: At startup time, we calibrate the threshold by measuring the LDR with the laser beam on and then set the threshold to 50% above that. Also, we ensure that the LDR reading is above the threshold for at least three consecutive measurements (approximately 300 ms) before the alarm is sent out to reduce false alarms due to ambient light changes.

Challenge 4: Pico W GPIO Current Limits

Problem: Pico W GPIOs have limited current supply capability, around 12 mA. This is not sufficient to power the buzzer and the multiple LEDs.

Solution: Use D400 NPN transistors to act as a switch between the GPIOs and the buzzer/LEDs. The GPIOs are connected to the transistor bases via 1 kΩ resistor. The transistors are used to supply the maximum 5 V to the buzzer and the LED arrays.

Challenge 5: PCB and Component Alignment

Problem: The alignment of the laser module and LDR to detect the beam passing through the gate structure is critical.

Solution: The custom-designed PCB has a gate shape with fixed points to mount the laser module and the LDR. The LDR is mounted on a header that is adjustable to make any corrections after assembling the system in case of misalignment.

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6. Component URLs Summary

Component	Vendor	URL
Raspberry Pi Pico W	Adafruit	https://www.adafruit.com/product/5526
Laser Module (KY-008)	Amazon	https://www.amazon.com/dp/B01MAWT5JX
LDR Sensor	Adafruit	https://www.adafruit.com/product/161
5V Active Buzzer	Amazon	https://www.amazon.com/dp/B0716GH3QD
5mm LED Pack (Red/Blue)	Amazon	https://www.amazon.com/dp/B01AUI4VQU
D400 Transistors	Amazon	https://www.amazon.com/dp/B07T5QRNXD
Resistor Kit (180Ω, 1kΩ, 10kΩ)	Amazon	https://www.amazon.com/dp/B08FD1XVL6
Custom PCB	JLCPCB	https://jlcpcb.com